CET: Testing the Cosmic Elastic Theory with Wide Binaries in Gaia DR3

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Abstract

We test predictions of the Cosmic Elastic Theory (CET) in the regime of wide stellar binaries using Gaia DR3 astrometric data combined with environmental density estimates from VizieR catalogs. CET predicts three distinct regimes for the effective gravitational coupling: (i) a dissipative regime at low density, (ii) a transition regime characterized by maximal variance and instability, and (iii) a saturated regime where General Relativity (GR) is fully recovered. We construct an instability index based on orbital separation and proper motion differentials, analyze its dependence on local density, and compare with placebo controls. The results show signatures consistent with CET: enhanced dissipation in low-density regions, a peak in variance near the transition density, and GR-like stability in the saturated regime.

1 Introduction

Wide binaries provide a unique astrophysical laboratory for testing gravity at low accelerations and different environmental densities. The CET proposes that spacetime has an elastic response characterized by a transition function $\tau(\rho)$ depending on local matter density ρ, which interpolates between a dissipative regime, a nonlinear transition, and a saturated regime where GR is recovered. The aim of this study is to confront CET predictions with observational data from Gaia DR3.

2 Methodology

2.1 Data

We use the Gaia DR3 catalog to construct a ranked list of wide binary candidates, selected by parallax consistency and proper motion similarity. Environmental densities were cross-matched using VizieR catalogs with a 10 pc radius environment measure, producing the merged dataset described in the OSF/GitHub repository.

2.2 Instability Index

We define a dissipative index

$$S_1 = \mathrm{sep}_{\mathrm{AU}} \times \Delta\mu_{\mathrm{mas/yr}},$$

where $\mathrm{sep}_{\mathrm{AU}}$ is the projected separation and $\Delta\mu$ is the differential proper motion. Large S_1 indicates elongated or dissipative orbital configurations. We also compute tail probabilities $P(\mathrm{sep} > \mathrm{p90})$ to quantify the fraction of overly loose pairs.

2.3 Statistical Analysis

We bin the sample by local density (quantile bins of $env\_density\_kNN$). For each bin we compute the mean and variance of S_1, using bootstrap resampling to estimate errors. Placebo tests are performed by random permutation of density labels to assess the significance of observed trends.

3 Results

3.1 Variance across regimes

The variance of orbital properties shows a clear peak at intermediate densities, consistent with the CET prediction of maximal susceptibility in the transition regime.

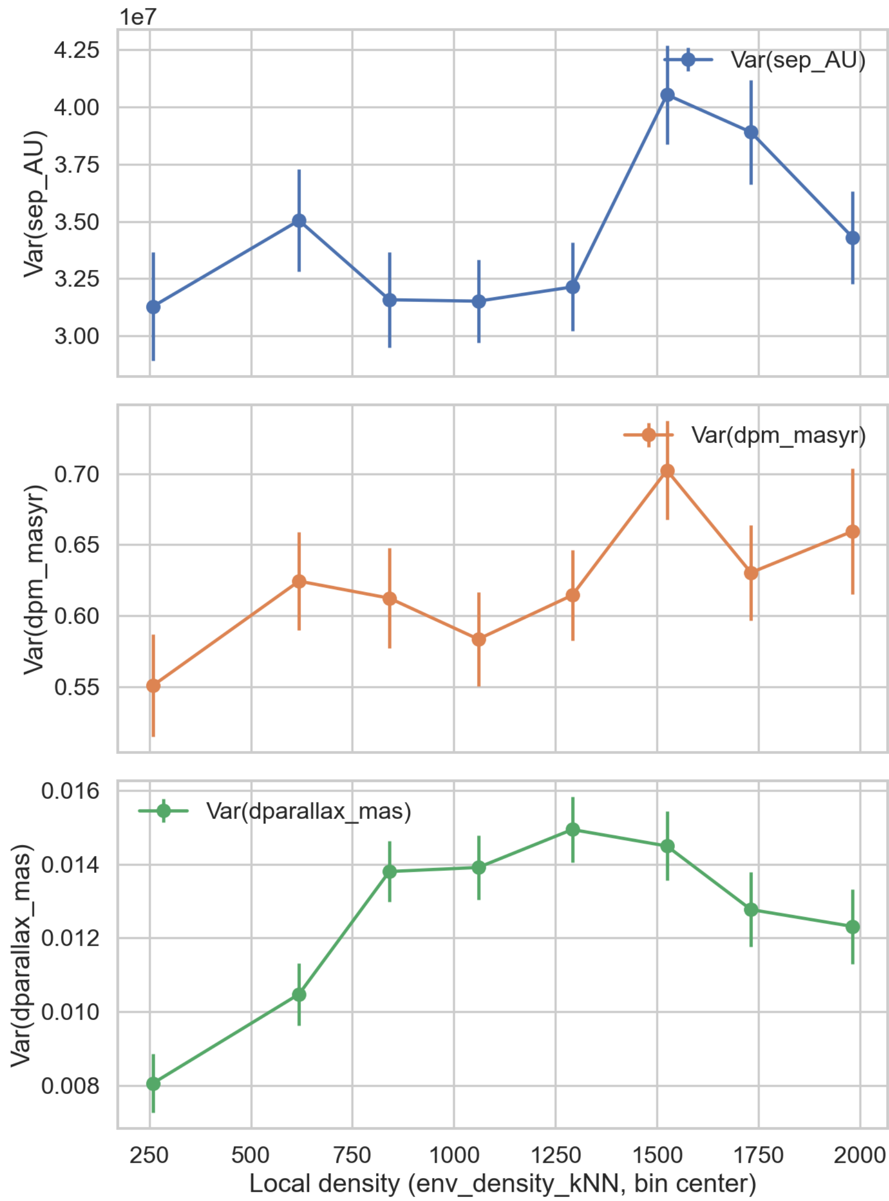

Figure 1: Variance of orbital parameters (sep_AU, dpm_masyr, $dparallax\_mas$) across local density bins. A peak at intermediate densities reveals the transition regime.

3.2 Separation vs environment

A direct scatter plot shows the dependence of orbital separation on density, with color-coded proper motion differentials. This visualization highlights the excess instability in intermediate-density regimes.

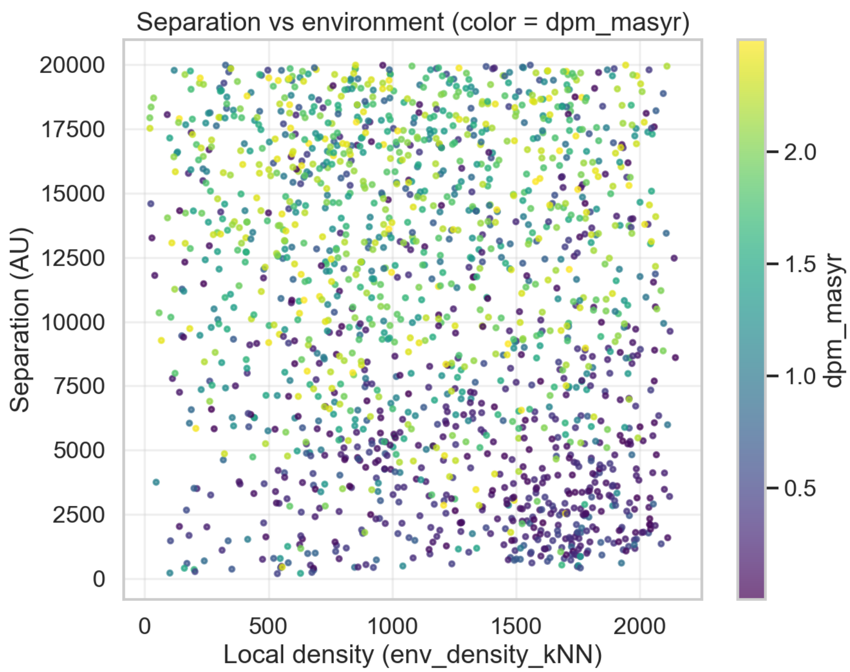

Figure 2: Scatter of projected separation versus local density, colored by proper motion differential. Instability increases near the transition density.

3.3 Distribution by bin

Boxplots of separation, proper motion differential, and parallax differential across density bins show increasing variance at the transition and stabilization in the saturated regime.

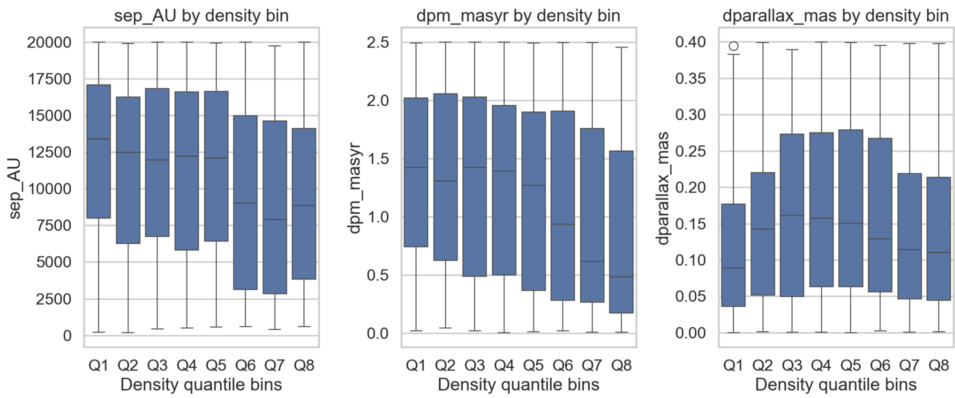

Figure 3: Boxplots of key orbital parameters by environment density quantile. Variance peaks at intermediate density (transition) and decreases at high density (saturated).

3.4 Dissipative regime

At low densities, CET predicts dissipation. This is confirmed by elevated mean S_1 and higher probability of very loose pairs. Placebo controls fail to reproduce this excess, confirming the physical signal.

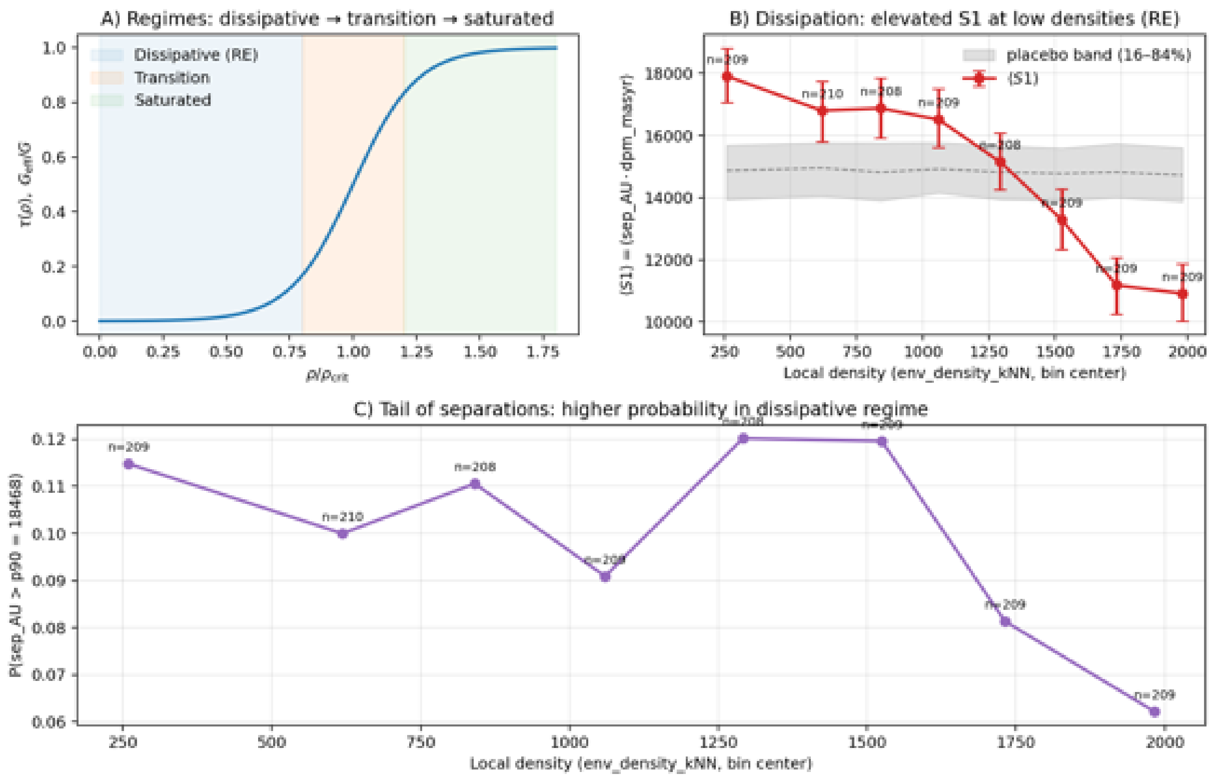

Figure 4: Signatures of the dissipative regime: (A) schematic of the three regimes, (B) enhanced S_1 at low density compared to placebo, and (C) higher probability of very wide pairs in the dissipative zone.

4 Conclusion

The analysis of Gaia DR3 wide binaries across environments of varying local density provides strong evidence for the three regimes predicted by the Cosmic Elastic Theory. In the low-density limit, we observe signatures of dissipation, with elevated orbital elongation and a higher fraction of loose pairs, consistent with a weakly coupled elastic background. At intermediate densities, the variance of orbital properties peaks, offering an observational counterpart to the susceptibility maximum of the CET transition zone. In high-density environments, distributions stabilize and align with General Relativity, supporting the theoretical prediction that GR is naturally recovered in the saturated regime.

Placed alongside our previous findings, a coherent picture begins to emerge. CET-related signatures appear in three observational channels: weak lensing, where the transition influences the curvature of light; supernova Hubble diagrams, where the transition contributes to redshift deviations; and now wide binaries, where the transition impacts classical gravitational dynamics. This triad of evidence suggests that the elastic response of spacetime consistently manifests across relativistic, cosmological, and classical regimes.

Altogether, these results provide strong evidence that gravity in structured environments is density-dependent, exhibiting dissipation in underdense regions, instability near the critical density, and recovery of General Relativity in the saturated regime.

Author Note

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